

COMS 4701 Artificial Intelligence

Homework 4 - Coding

Due date: November 26, 2024

Part I: Classification

In this problem, you will use various classifiers in the machine learning **sklearn** package to learn classification models for a chessboard-like dataset. In your starter code, you will find [input.csv](#), containing a set of data points.

- (a) Open the dataset in python. Make a scatter plot of the dataset showing the two classes with two different patterns.
- (b) Split your data into training (60%) and testing (40%).
- (c) Use the following methods to build a classifier.
 1. **k-nearest Neighbors.** Try `n_neighbors = [1, 3, 5, ..., 19]` and `leaf_size = [5, 10, 15, ..., 30]`.
 2. **Logistic Regression.** Try values of `C = [0.1, 0.5, 1, 5, 10, 50, 100]`.
 3. **Decision Trees.** Try values of `max_depth = [1, 2, 3, ..., 50]` and `min_samples_split = [2, 3, ..., 10]`.
 4. **Random Forest.** Try values of `max_depth = [1, 2, 3, 4, 5]` and `min_samples_split = [2, 3, ..., 10]`.
 5. **Adaboost.** Try `number_of_estimators = [10, 20, ..., 70]`.

Make sure to use five fold cross-validation. Train-test splitting and cross-validation are both functionalities readily available in sklearn.

For each method, search for a good setting of the parameters with to obtain high classification accuracy. You can use sklearn's GridSearchCV to help with this.

After finding the optimal parameter values, using the training data, record the corresponding best training accuracy. Then apply the best model to the testing data and record the actual test accuracy. Both scores should be numbers between zero and one.

From your code, add all your results to `output.csv`.

To visualize the result of each case of classification method, plot the decision boundary for the best parameters. You should use matplotlib to output an image containing the data points and boundaries of each method for the test set. An example is provided below. There is code written for this in your starter code. Discuss which method performs the best.

What To Submit:

- (30 points): Your code [classifiers.py](#) which will consist of the code for your classifiers. There are many ways to implement these classifiers, and we will allow for small variations.
- (10 points): Your [output.csv](#) which consists the performance of your models, as described above.
- (20 points): Your [PDF report](#) consisting of your plots along with a brief analysis of the performance of each method.

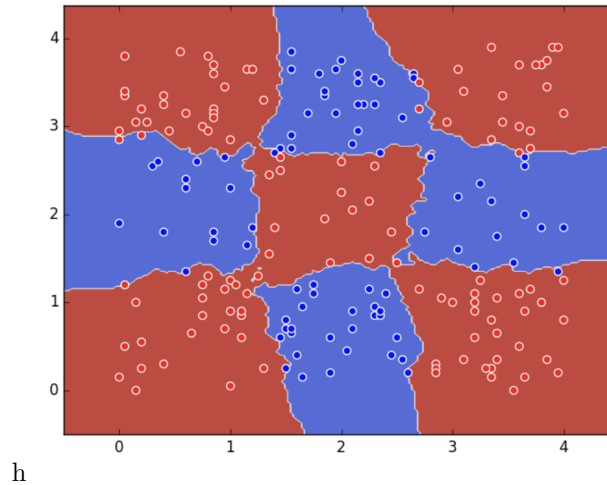


Figure 1: Example Plot

Part II - High dimensionality: Curse or Blessing?

In this question we will study the curse of dimensionality. To witness the curse, consider the following K-Nearest Neighbors (KNN) query experiment:

- Use sklearn to generate a dataset from a standard Gaussian distribution. Then select a random query point and calculate the DMAX/DMIN ratio (the ratio of the distance of furthest point from the query to distance of nearest point to the query).
- Repeat part (a) for different values of dimensionality of the data, from $d = 1$ to $d = 100$. Plot these results and report your findings.
- Research suggests that high dimensionality is rather a blessing. In this question, you will conduct some research on whether high dimensionality can be a blessing. Find some research paper(s) that tackles this issue and summarize your findings. Please write an argument about this topic in up to one page. Cite all your references.

Suggested reading for this question (feel free to dig in more references):

- Wikipedia page on the curse/blessing of dimensionality.
- Courseworks article: a few useful things to know about machine learning.

What To Submit:

- (20 points): Your code `curse.py` which will consist of your code to produce the plot in part (b).
- (20 points): Add to the PDF report from part I, the following elements:
 - (5 points) The plot from part (b).
 - (15 points): Your research about the blessing of high dimensions.